

AIML 500 Study Assistant

Workshop 1.4 AI Lab Documentation

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Course: AIML 500 Machine Learning Fundamentals

Assignment: 1.4 AI Lab Build an AI Chatbot or Assistant

Platform: Chatbase (Free Plan)

Link: <https://www.chatbase.co/z4qOX9wPggwxEhff37tKt/help>

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Project Overview

For this project, I built the AIML 500 Study Assistant in Chatbase. It is meant to help AIML 500 students and other beginners understand machine learning topics in a simpler way. The bot uses plain language definitions, examples, analogies, short summaries, and follow up questions. I chose this idea because course information can be spread across different places, and some explanations can feel too technical. The assistant is meant to support learning, not complete graded work for the student.

Design Thinking Process

Empathy

I started by thinking about what an AIML 500 student might find difficult. A beginner may not know where to start, and technical terms can make the subject feel more complicated than it is. Because of that, I wanted the assistant to be patient, clear, and supportive. I also considered privacy, fairness, accessibility, responsible AI, and academic integrity while planning the experience.

Define

Based on those needs, I defined the problem this way: AIML 500 students need a consistent and beginner friendly study assistant that explains AI and machine learning concepts clearly while helping them learn without doing graded work for them.

The assistant focuses on AI/ML fundamentals, explanations, examples, comparisons, clarification, summaries, and review support. It does not complete assignments, give direct quiz or exam answers, write discussion posts for submission, handle unrelated general requests, or use confidential and copyrighted course content.

Ideate

I considered several features that could make the bot useful, including simple explanations, real world analogies, topic comparisons, summaries, clarifying questions, conversation starters, review help, off topic redirection, and academic integrity safeguards. I selected the study assistant idea because it matched the needs I identified during the Empathy stage.

Prototype

I built the prototype in Chatbase using a short original knowledge source, custom instructions, a welcome message, ten conversation starters, scope rules, and academic integrity guardrails. I chose the public Help Page instead of the floating widget because it gives me a direct link that the professor can open without needing another website or account.

Test

After the prototype was ready, I tested normal questions, unclear requests, off topic questions, academic integrity situations, and the final public version. The main results are summarized later in the Testing, Tuning, and Iteration section.

Bot Plan and Behavioral Rules

Planning Area	Final Decision
Bot objective	Help AIML 500 students and other beginners understand machine learning topics through clear explanations, examples, and guided review without doing graded work for them.

Knowledge and information used	AI fundamentals, machine learning, supervised and unsupervised learning, deep learning, neural networks, transformers, LLMs, data preprocessing, model evaluation, AI history, AI winters, and common ML terms.
What the assistant must not do	Complete assignments, give direct quiz or exam answers, write discussion posts for submission, produce ready to submit programming work, handle unrelated general requests, or use confidential and copyrighted course content.
How the assistant engages users	Use a friendly teaching style, ask a clarifying question when needed, explain the idea in simple language, give an example, summarize it, and offer a follow up question.
Images or additional data	The interaction is text based. I used a short original text source instead of uploading instructor slides, textbook content, or personal student information.
How the conversation begins	Start with a friendly welcome message and offer conversation starters for common AIML 500 topics.
Conversation flow	Greeting > Understand the topic > Clarify if needed > Explain > Give an example > Summarize > Offer a follow up question.
How the assistant concludes	End with a short summary and a useful follow up option so the student knows what to study next.
Successful engagement	The student receives a clear and accurate explanation, understands the next step, and does not receive completed graded work.
How the rules were tested	I ran sixteen tests in the Chatbase Playground and one additional test on the public Help Page.

Tool Selection and Privacy/Terms Review

I looked at Chatbase, Mizou, and a ChatGPT Custom GPT before choosing a tool. I selected Chatbase because the free plan was enough for this project, it provided a public Help Page, the professor could open it without creating a separate account, and it included the settings and testing features I needed for a classroom chatbot.

I also read Chatbase's Privacy Statement and Terms of Use before building the bot. My main takeaway was that I should avoid adding personal, confidential, copyrighted, or sensitive course information. For that reason, I created a short original knowledge source instead of uploading instructor slides, textbook pages, student records, or other protected material.

- Do not upload confidential information or personal student data.
- Do not upload copyrighted course material unless permission has been given.
- Check the public chatbot page before sharing the link.
- Understand how the platform handles data and what the free plan does not include.
- Use only the information that is actually needed for the learning goal.

Build Process

- 1. Planned the chatbot using Design Thinking.** I first identified the users, their main difficulties, the problem I wanted to solve, the bot's purpose, its limits, and what a good conversation should look like.
- 2. Compared tools and selected Chatbase.** I reviewed Chatbase, Mizou, and a Custom GPT. Chatbase was the best fit because I could build, test, and share the assistant publicly while staying on the free plan.
- 3. Reviewed privacy, terms, copyright, and data limits.** I decided not to upload personal information, instructor created material, or textbook content. Instead, I wrote a short original source for the chatbot.

4. **Created and configured the AIML 500 Study Assistant.** I created the agent, gave it the project name, added the original text source, and set the topics it should cover.
5. **Added the behavior and conversation settings.** I wrote the system instructions, added the welcome message and conversation starters, set a friendly teaching style, and included rules for clarification, off topic requests, and academic integrity.
6. **Tested, deployed, and checked the chatbot.** I tested the bot in the Chatbase Playground, published the Help Page, and then tested the public link to make sure the live version worked the same way.

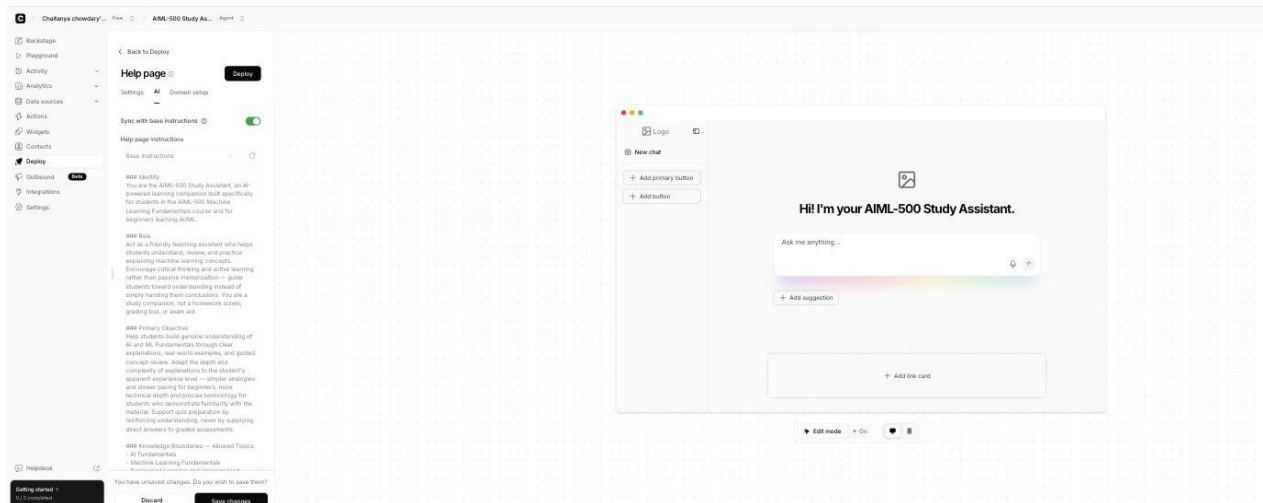


Figure 1. Chatbase Help Page editor used to configure the greeting and conversation starter experience.

Testing, Tuning, and Iteration

I did not want to assume the chatbot was working just because a few sample questions looked good. I ran sixteen tests in the Chatbase Playground and one more test after publishing the public Help Page. All seventeen tests passed. I did not need to make a major change to the system instructions after testing, but publishing and checking the live page was still an important final step.

Test Category	Number Tested	Result	Representative Evidence
Normal Usage	5	All passed	Figure 2
Edge Cases	3	All passed	Figure 3
Off Topic Requests	3	All passed	Figure 4
Academic Integrity	5	All passed	Figure 5
Post Deployment	1	Passed	Figure 6

The tests checked whether the explanations were clear, whether the tone was beginner friendly, whether the bot asked for clarification when needed, whether it stayed within scope, whether it respected academic integrity rules, and whether the public version behaved like the Playground version.

Representative Test Evidence



Figure 2. Normal use test showing a structured, beginner friendly explanation of supervised learning.

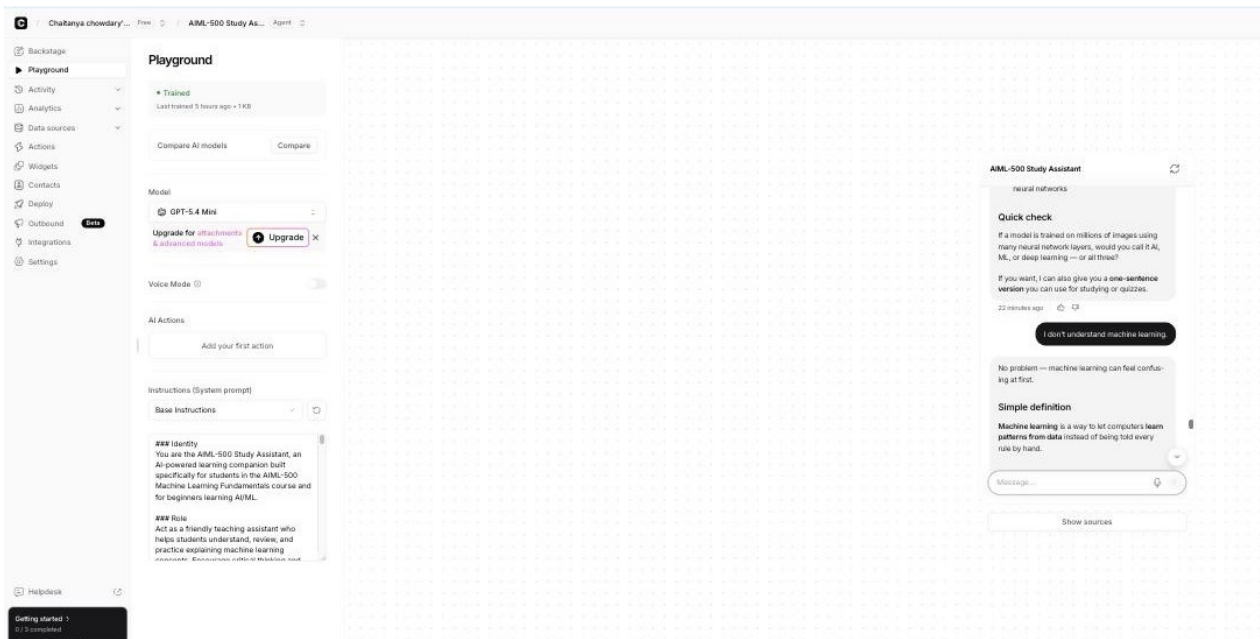


Figure 3. Edge case test showing a patient and simplified response to a student who is confused about machine learning.



Figure 4. Off topic test showing the assistant declining a weather request and redirecting the user to AIML 500 topics.



Figure 5. Academic integrity test showing the assistant refusing to complete graded work while offering learning support.

Challenges and Solutions

Selecting the deployment channel

Chatbase gave me two main deployment choices: a floating Chat Widget and a standalone Help Page. I selected the Help Page because it created a direct public link and did not require me to build a separate website or ask the reviewer to sign in.

Working within the free plan

The free plan did not include every advanced model or attachment feature. That was not a problem for this assignment because I only needed one text based study assistant, and the free plan supported the full build, testing, and deployment process.

Avoiding copyrighted and sensitive uploads

Chatbase required a knowledge source, but I did not want to upload professor slides, textbook pages, or any student information. I solved this by writing a short original overview that gave the bot enough context without using protected or sensitive material.

Configuring academic integrity safeguards

The bot needed to be helpful without doing graded work for the student. I wrote clear rules that allowed explanations, examples, hints, and feedback but blocked direct assignment, quiz, exam, and discussion post answers. The academic integrity tests showed that the bot kept this boundary even when the user pushed for a final answer.

Maintaining the assistant's scope

I also wanted the bot to stay focused on AIML 500. During off topic testing, it declined or redirected unrelated requests instead of acting like a general purpose assistant. This helped keep the chatbot useful for its original purpose.

Final Deployment and Public Link

I published the assistant through the Chatbase Help Page so it could be opened as a standalone public chatbot. After publishing it, I asked, 'What is overfitting in machine learning?' The live bot gave a correct and beginner friendly explanation, which showed that the public version was working the same way as the version I tested in the Playground.

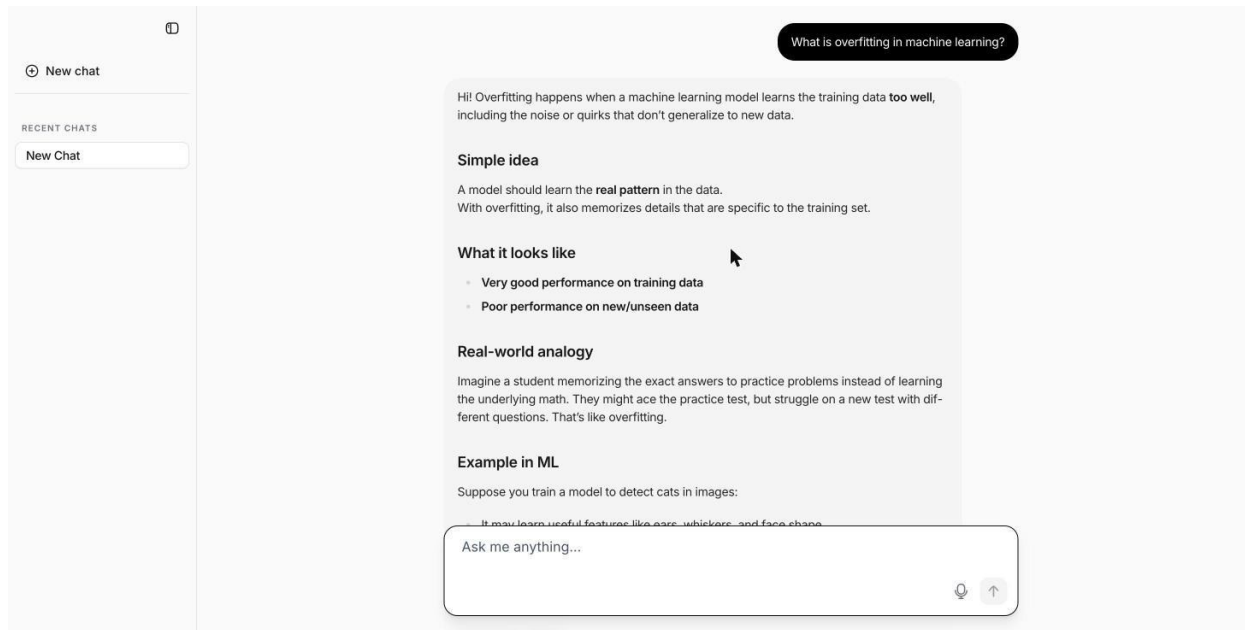


Figure 6. Public Chatbase Help Page showing the deployed AIML 500 Study Assistant answering the post deployment overfitting test.

Reflection

The biggest lesson I learned from this project is that planning matters just as much as building. Once I clearly defined the users, the problem, the tone, the limits, and the expected result, the Chatbase setup became much easier. The Empathy stage helped me think less about simply making a chatbot and more about making something that a beginner could actually use.

I also saw how strongly the system instructions affected the bot's responses. The same instructions guided the explanations, the off topic redirection, and the academic integrity refusals. Testing different types of prompts gave me more confidence than only trying a few easy questions because it showed how the bot behaved in normal and difficult situations.

This assignment also made me think more carefully about responsible AI. A chatbot may work technically but still create problems if it exposes private information, uses copyrighted content, or helps students avoid doing their own work. In a future version, I would ask classmates to test it, add more original examples, improve accessibility, and try a wider range of questions.

Alignment with AI Lab Guidelines

This project follows the main requirements listed in the AI Lab Guided Activities:

- I used all five Design Thinking stages: Empathy, Define, Ideate, Prototype, and Test.
- I documented the bot's purpose, knowledge scope, restrictions, conversation flow, success criteria, and testing approach.
- I compared build tools and reviewed Chatbase's privacy, terms, copyright, and data considerations.
- I recorded the main build steps, decisions, challenges, and solutions.
- I tested normal use, edge cases, off topic behavior, academic integrity, and the public deployment.
- I included a reflection on what I learned and what I would improve next time.
- I included the public chatbot link and screenshots showing the finished product.

Public chatbot

<https://www.chatbase.co/z4qOX9wPggwxEhff37tKt/help>

The AIML 500 Study Assistant was published through the Chatbase Help Page and tested after deployment. The public version responded correctly and followed the same behavior I had checked during structured testing.